120|

$$f(x) = \begin{cases} 0, & x<0 \\ ax, & 0 \le x < 2 \\ 0, & x > 2 \end{cases}$$

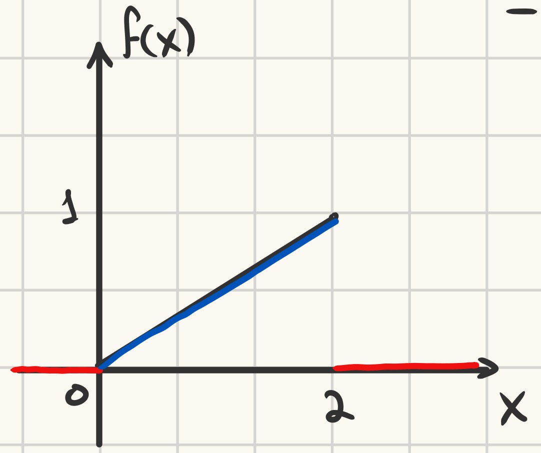
Условие нормировки: $\int_{\mathbb{R}} f(x)\,dx = 1$ $\quad \int_0^2 ax\,dx = a\frac{x^2}{2}\Big|_0^2 \Rightarrow 2a = 1 \Rightarrow a = \frac{1}{2}$

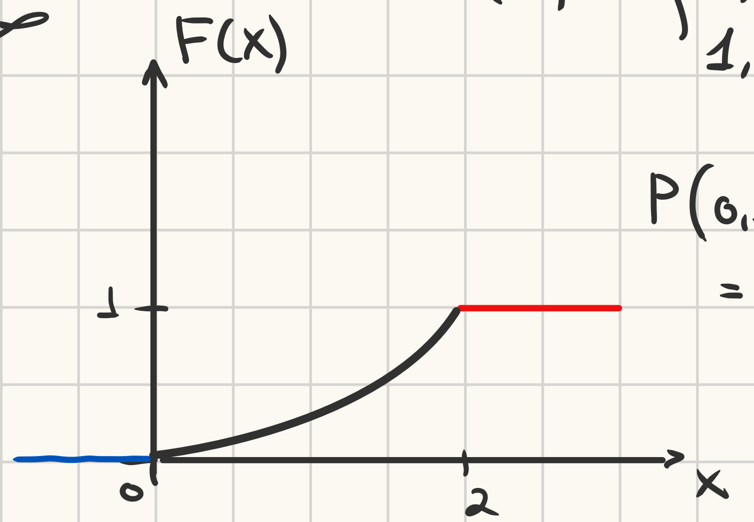
2. $E\xi = \int_{\mathbb{R}} f(x)\, x\, dx = \int_0^2 x \cdot \frac{1}{2} x = \frac{x^3}{6}\Big|_0^2 = \frac{4}{3}$

3. $D\xi = E\xi^2 - (E\xi)^2 = \int_0^2 x^2 \cdot \frac{1}{2} x\, dx - \left(\frac{4}{3}\right)^2 = \frac{x^4}{8} - \frac{16}{9} = \frac{2}{9}$

$\sigma = \sqrt{D\xi} = \sqrt{\frac{2}{9}} \approx 0{,}47$ $\qquad \int_0^x \frac{1}{2} x\,dx = \frac{x^2}{4}\Big|_0^x = x^2/4$

$F(x) = P(\xi < x) = \int_{-\infty}^{x} f(x)\,dx$ $\qquad F(x) = \begin{cases} 0, & x<0 \\ x^2/4, & 0 \le x \le 2 \\ 1, & x > 2 \end{cases}$




$P(0{,}5 < \xi < 1{,}5) = F_\xi(1{,}5) - F_\xi(0{,}5) = \frac{(1{,}5)^2 - (0{,}5)^2}{4} = \frac{1}{2}$

121|

$$f(x) = \begin{cases} 0, & x < \pi/2 \\ a \cdot \cos(x), & -\pi/2 < x < \pi/2 \\ 0, & x > \pi/2 \end{cases}$$

$\int_{-\pi/2}^{\pi/2} a\cos x\,dx = 1 \qquad a\sin x\Big|_{-\frac{\pi}{2}}^{\pi/2} = a + a = 2a = 1 \Rightarrow a = \frac{1}{2}$

$$E\xi = \int_{-\pi/2}^{\pi/2} x\,\frac{1}{2}\cdot\cos(x)\,dx = \begin{bmatrix} u = x \\ dv = \cos x\,dx \\ du = dx \\ v = \sin(x) \end{bmatrix} = \frac{1}{2}\left(x\sin x\Big|_{-\frac{\pi}{2}}^{\pi/2} - \int_{-\frac{\pi}{2}}^{\pi/2} \sin x\,dx\right) =$$

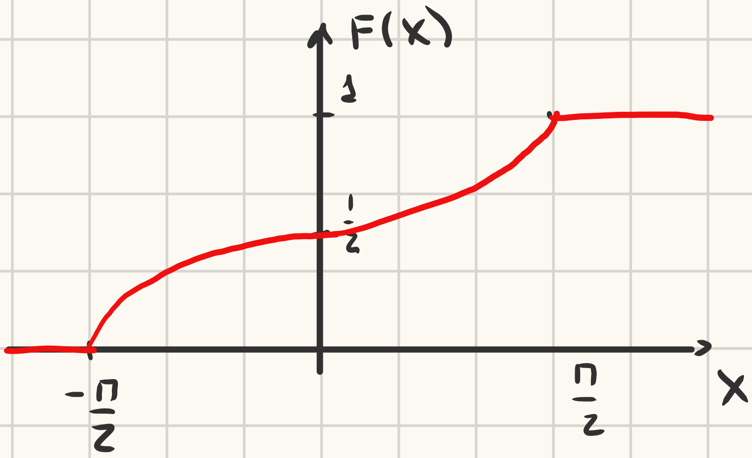
$$= \frac{1}{2}\left(\frac{\pi}{2} - \frac{\pi}{2} + \cos(x)\Big|_{-\frac{\pi}{2}}^{\pi/2}\right) = 0$$

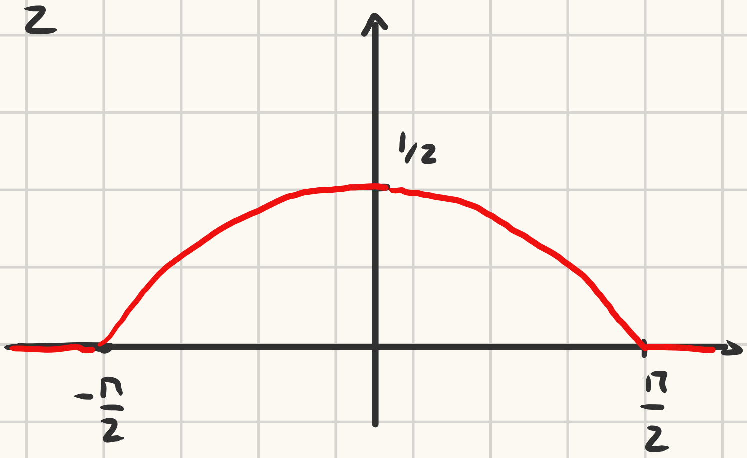
$$D\xi = E(\xi)^2 - (E\xi)^2 = \int_{-\frac{\pi}{2}}^{\pi/2} x^2 \frac{1}{2}\cos x\,dx = \begin{bmatrix} u = x^2 & du = 2x\,dx \\ dv = \cos x\,dx & v = \sin(x) \end{bmatrix} =$$

$$= \frac{1}{2}\left(x^2 \sin(x)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \int_{-\frac{\pi}{2}}^{\pi/2} 2x\sin x\,dx = \begin{bmatrix} u = 2x & du = 2\,dx \\ dv = \sin x\,dx & v = -\cos x \end{bmatrix} =\right.$$

$$= \frac{1}{2}\left(\frac{\pi^2}{4} + \frac{\pi^2}{4} - 2x\cos x\Big|_{-\frac{\pi}{2}}^{\pi/2} - \int_{-\frac{\pi}{2}}^{\pi/2} 2\cos x\,dx\right) = \frac{\pi^2}{4} - 2 = \frac{\pi^2 - 8}{4}$$

$$F(x) = \begin{cases} 0, & x < -\pi/2 \\ \frac{1}{2}\sin x + \frac{1}{2}, & -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ 1, & x > \pi/2 \end{cases} \qquad \int_{-\frac{\pi}{2}}^{x} \frac{1}{2}\cos x\,dx = \frac{1}{2}\sin x\Big|_{-\frac{\pi}{2}}^{x} = \frac{1}{2}\sin x + \frac{1}{2}$$




$$P\left(-\frac{\pi}{6} < \xi < \frac{\pi}{6}\right) = P\left(\frac{\pi}{6}\right) - P\left(-\frac{\pi}{6}\right) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$P\left(-\pi < \xi < \frac{\pi}{2}\right) = P\left(\frac{\pi}{2}\right) - P(-\pi) = 1$$

<u>122|</u>

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{a}{x^4}, & x \ge 1 \end{cases}$$

$$\int_1^{+\infty} \frac{a}{x^4}\,dx = 1 = \frac{a}{(-3)} x^{-3}\Big|_1^{+\infty} = 1 \Rightarrow \lim_{\beta \to +\infty} \frac{a}{(-3)}\beta^{-3} + \frac{a}{3} 1^{-3} = 0 + \frac{a}{3} = 1 \Rightarrow a = 3$$

$$E\xi = \int_{\mathbb{R}} \frac{3}{x^4} x\,dx = \int_1^{+\infty} \frac{3}{x^3}\,dx = 0 + \frac{3}{2} 1^{-2} = \frac{3}{2} = 1{,}5$$

$$D\xi = \int_1^{+\infty} \frac{3}{x^4} x^2\,dx - 1{,}5^2 = \int_1^{+\infty} \frac{3}{x^2}\,dx - 1{,}5^2 = 3 - \frac{9}{4} = \frac{3}{4} \qquad \sigma_\xi = \frac{\sqrt{3}}{2}$$

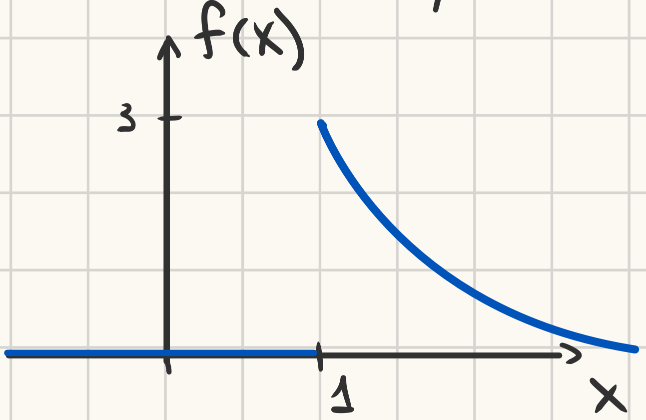
$$F(x) = P(\xi < x) = \int_{-\infty}^{x} f(x)\,dx \qquad F(x) = \begin{cases} 0, & x < 1 \\ 1 - \frac{1}{x^3}, & x \ge 1 \end{cases}$$

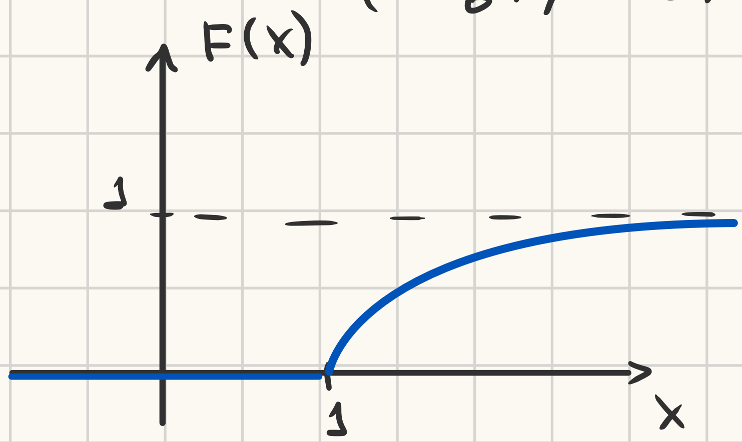
$$\int_1^x \frac{3}{x^4}dx = \frac{3}{(-3)}x^{-3} - \frac{3}{(-3)}1^3 = 1-\frac{1}{x^3}$$

$$P(2<\xi<4) = F(4)-F(2) = 1-\frac{1}{64}-\left(1-\frac{1}{6}\right) = \\ = 1/8 - 1/64 = 7/64$$

$$P(0<\xi<3) = F(3)-F(0) = 1-\frac{1}{27}-0 = \frac{26}{27}$$

$$P(4<\xi) = 1-p(\xi\leq 4) = 1-F(4) = 1-\left(1-\frac{1}{64}\right) = \frac{1}{64}$$

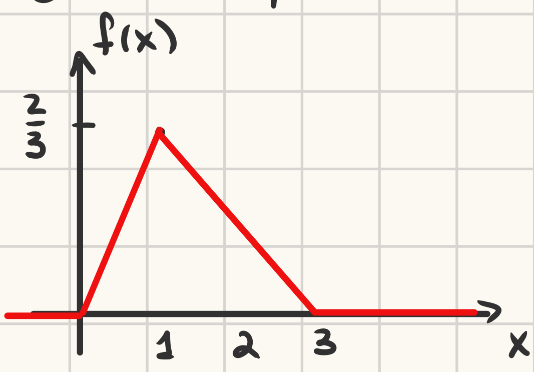


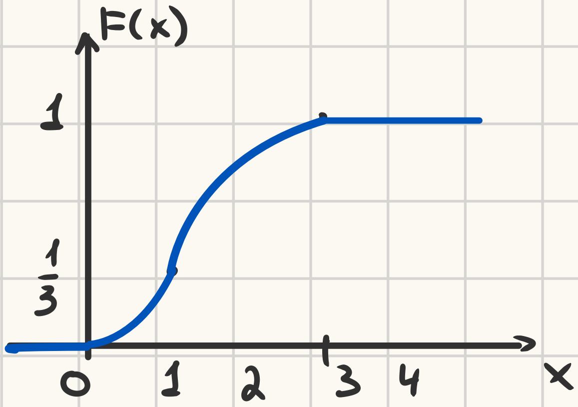




**125**

$$f(x) = \begin{cases} 0, & x<0 \\ ax, & 0\leq x\leq 1 \\ \frac{a}{2}(3-x), & 1<x\leq 3 \\ 0, & x>3 \end{cases}$$

$$\int_0^1 ax\,dx + \int_1^3 \frac{a}{2}(3-x)\,dx = 1. \qquad \frac{ax^2}{2}\Big|_0^1 + \frac{a}{2}\left(3x-\frac{x^2}{2}\right)\Big|_1^3 = \frac{3a}{2} \Rightarrow a = \frac{2}{3}$$




$$F(x) = \begin{cases} 0, & x<0 \\ x^2/3, & 0\leq x\leq 1 \\ x-\frac{x^2}{6}-\frac{5}{6}+\frac{1}{3}, & 1<x\leq 3 \\ 1, & x>3 \end{cases}$$

$$\int_1^x 1-\frac{x}{3} = x-\frac{x^2}{6}\Big|_1^x = x-\frac{x^2}{6}-\frac{5}{6}$$

$$P(0{,}5<\xi<2) = F(2)-F(0{,}5) = 2-\frac{4}{6}-\frac{5}{6}+\frac{1}{3}-\frac{1}{12} = \frac{1}{2}+\frac{1}{3}-\frac{1}{12} = \frac{3}{4}$$

$$E\xi = \int_0^1 \frac{2}{3}x \cdot x\,dx + \int_1^3 \left(1-\frac{x}{3}\right)x\,dx = \frac{2}{9}x^3\Big|_0^1 + \frac{x^2}{2} - \frac{x^3}{9}\Big|_1^3 = \frac{4}{3}$$

$$D\xi = \int_0^1 \frac{2}{3}x \cdot x^2\,dx + \int_1^3 \left(x-\frac{x^2}{3}\right)x^2\,dx - \frac{16}{9} = \frac{1}{6}x^4\Big|_0^1 + \frac{x^4}{4} - \frac{x^5}{15}\Big|_1^3 - \frac{16}{9} = \frac{7}{18}$$

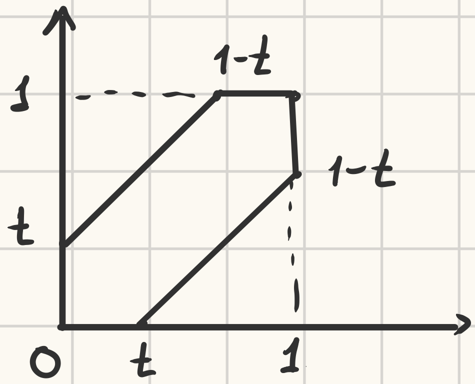
$$\sigma_\xi = \frac{\sqrt{14}}{6}$$

<u>131|</u> $0 \le x, y \le 1$

$y$ – время прихода 2^го

$x$ – время прихода 1^го

$\xi = |x-y| \in [0;1]$




$$F_\xi(t) = P(\xi < t) = \frac{S_\xi(A)}{S_A} = \frac{1-(1-t)^2 \cdot 2 \cdot \frac{1}{2}}{1} = 2t - t^2$$

$t \in [0;1]$

$$f_\xi(t) = F'_\xi(t) = 2-2t,\quad t \in [0;1]$$

$$E\xi = \int_0^1 t(2-2t)\,dt = t^2 - \frac{2}{3}t^3\Big|_0^1 = \left(1-\frac{2}{3}\right) - 0 = 1/3 \text{ часа} = 20 \text{ минут}$$

$$D\xi = \int_0^1 t^2(2-2t)\,dt - \left(\frac{1}{3}\right)^2 = \frac{2}{3}t^3 - \frac{t^4}{2}\Big|_0^1 - \frac{1}{9} = \frac{1}{18} \text{ часа}^2$$

$\sigma \approx 14$ минут